

EDGAR-X: A Point-in-Time Benchmark for Bankruptcy Prediction from SEC Filings

What a Filer Knew, and What a Model Is Told:
Filing Text, Financial Ratios, and Four Channels of Information-Set Contamination

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Abstract

A model that predicts a bankruptcy in 2015 should see only what was on file in 2015. Historical bankruptcy datasets can violate that constraint through later-restated financials, present-day classifications, survivor-defined universes, or evaluation schemes that expose future observations during training. We introduce **EDGAR-X**, a benchmark for predicting US Chapter 7 or Chapter 11 bankruptcy within 365 days of a Form 10-K, in which every input is fixed against that filing’s EDGAR acceptance timestamp: as-filed financial facts, the SIC code carried on the filing, CIK identity, and the actual petition date as event time. The panel holds **59,500 10-K observations from 2012–2024 carrying 437 bankruptcy-positive observations**, of which 54,351 observations and 408 positive observations are testable under rolling temporal evaluation. One bankruptcy event can label more than one 10-K, so positive observations are counted separately from distinct events throughout.

We put the benchmark to two uses. First, we audit four ways of violating the historical information set, with signed effects and decision rules fixed before measurement. The channels do not behave alike: one produces a negative effect, one changes the composition of the benchmark rather than its scores, one lacks the positive-class exposure to be testable at all, and one is a heterogeneous protocol sensitivity that firm-grouped cross-validation does not remove. Contamination is not a single quantity waiting to be subtracted.

Second, we compare modalities under a strict information set. Filing text outperforms financial ratios by a wide margin: event-weighted average precision **0.1301 against 0.0425**, a **206% relative improvement, positive in 12 of 12 cohorts**. Combining both modalities reaches **0.1447**, but ratios add only a modest and inconsistent increment over text alone. Under the common sparse linear model family studied here, filing text accounts for most of the predictive performance observed relative to the ratio-only baseline.

1 Introduction

A prediction made in March 2015 can only use what existed in March 2015. This is obvious when stated and difficult to honour in practice, because the data a researcher downloads is a present-day object. Financial statements carry values that were revised years after they were first filed. Industry classifications reflect the code a registrant carries today. Identity maps enumerate the firms that still exist, which for a bankruptcy study is a selection on the outcome variable. Each

of these substitutions quietly moves information backwards in time, and a model trained on the result is answering a question no practitioner could have asked.

For bankruptcy prediction the stakes are structural rather than incidental. The positive class is rare (roughly seven in a thousand filings here), and it is precisely the failing registrants whose records behave differently: they stop filing, so they stop being restated; they disappear from present-day universes; they file late, so filing date and acceptance date diverge. Any convention that treats the surviving firm as the typical firm is therefore correlated with the label. That correlation is what makes an information-set violation a measurement problem and not a tidiness problem.

The usual response is to assert that such contamination inflates measured performance, and to treat point-in-time (PIT) reconstruction as a correction of known sign and roughly known size. We find that framing too coarse to survive measurement. Auditing four contamination channels on one panel, with signed effects and decision rules fixed in advance, we find that the channels differ from one another in direction, in magnitude, and in the estimand that it even makes sense to compute. One produces a negative effect. One is not a performance effect at all. One cannot be tested here for lack of positive-class exposure. One is a protocol sensitivity that firm-grouped cross-validation does not address on this benchmark.

What EDGAR-X reconstructs. EDGAR-X fixes the information set to a single clock: the EDGAR acceptance timestamp of the 10-K being scored. A financial fact enters an observation only if it comes from a submission accepted at or before that timestamp. Industry classification is the SIC code on the filing itself. Identity is the CIK printed on the document. The event time is the actual bankruptcy petition date, not the date it was disclosed. Training rows come only from cohorts strictly earlier than the cohort being tested. Figure 1(a) states the rule; Figure 1(b) shows the pipeline that enforces it. Each requirement is enforced by a named module and checked in continuous integration.

Research questions. The benchmark exists to answer three questions.

RQ1. How do different violations of historical information integrity affect a bankruptcy benchmark, in direction, in magnitude, and in the quantity that can be estimated at all?

RQ2. Under strict PIT evaluation, how much predictive signal is carried by structured financial ratios against 10-K narrative text?

RQ3. Once filing text is available, do financial ratios add meaningful incremental information?

Findings. On RQ1, the four channels behave differently enough that a single “cost of contamination” would be misleading; we report them separately and decline to aggregate incommensurable estimands. On RQ2, the modality effect is the largest and most robust result in the study: filing text raises event-weighted average precision from 0.0425 to 0.1301, a 206% relative gain that is positive in every one of the twelve test cohorts and survives both frozen sensitivities and every single-cohort deletion. On RQ3, the answer is a qualified no: adding ratios to text improves the event-weighted aggregate by 11.2%, but only three of twelve cohort intervals exclude zero on the positive side and one cohort is significantly negative.

Contributions.

1. **A point-in-time bankruptcy benchmark with an enforced information-set contract.** 59,500 Form 10-K observations over 2012–2024 carrying 437 positive observations, with 54,351 observations and 408 positive observations testable under rolling evaluation. The

contract is falsifiable: every rule in §3.2 names the module that enforces it, and a violation is a test failure rather than a matter of interpretation.

2. **A signed audit of four contamination channels, each specified and frozen before measurement, showing that their effects differ in direction, magnitude and estimand.** Each channel carries a declared estimand, a decision rule fixed before measurement, and a minimum detectable effect, so that null results are reportable rather than invisible.
3. **A controlled PIT comparison of filing text against financial ratios,** scored on identical test rows under one protocol and one model family, which finds that text carries most of the signal and that the multimodal increment over text is modest and heterogeneous.
4. **An evaluation design calibrated to a rare positive class.** Average precision under a paired CIK-cluster bootstrap, four aggregation views reported together rather than one chosen after the fact, and an explicit minimum detectable effect of roughly 24–33% relative, which is why several audit results are reported as bounds on dominance rather than as evidence of absence.

A power constraint runs through the paper. This panel resolves a 200% modality effect comfortably and cannot resolve a 10% contamination effect at all. We state that asymmetry wherever it bears on a conclusion.

2 Related Work

2.1 Classical bankruptcy prediction

The quantitative literature begins with accounting ratios. Beaver (1966) showed that ratios computed before failure separate failing from non-failing firms, and Altman (1968) combined them into a multiple discriminant function that became the field’s reference model. Ohlson (1980) replaced discriminant analysis with a logit specification, relaxing its distributional assumptions and returning a probability rather than a score.

The more consequential shift for this paper is temporal. Shumway (2001) argued that single-period models are misspecified for an outcome that unfolds over time, and proposed a discrete-time hazard model in which each firm-year is an observation with its own forward horizon. Chava and Jarrow (2004) extended hazard estimation with industry effects, and Campbell et al. (2008) estimated a dynamic panel specification combining accounting and market variables. EDGAR-X inherits this observational convention, a filing-level unit with a forward horizon, and adds the constraint that the unit’s information set be the one that existed at that moment.

2.2 Machine learning for financial distress

Ratio-based prediction has been revisited with modern learners. Barboza et al. (2017) compare machine-learning models against the discriminant and logit baselines on accounting ratios, and Mai et al. (2019) move past ratios entirely, using deep models over textual disclosures and reporting that text contributes predictive signal.

This paper is deliberately conservative in model family, using sparse linear models over TF-IDF so that the information-set contract stays auditable end to end. That is a scoping decision, not a claim about what stronger architectures would achieve; the protocol question we study is orthogonal to model capacity.

2.3 Textual information from SEC filings

The case for treating filing text as data rests on a finding about language rather than about models. Loughran and McDonald (2011) showed that general-purpose sentiment dictionaries misclassify financial prose, since words that read as negative in ordinary usage are routine in a 10-K, and built finance-specific word lists in response. Li (2008) established annual-report readability as an

informative attribute of filing text, and Campbell et al. (2014) examined the information content of mandatory risk-factor disclosures in corporate filings, which is one reason Item 1A and Item 7 are the two sections this benchmark extracts. Loughran and McDonald (2016) survey the broader literature.

Our text arm uses a plain sparse representation rather than a finance-specific lexicon or a learned embedding, again for reproducibility rather than because we expect it to be the strongest available choice.

2.4 Evaluation integrity

Concern about the information set is old in empirical finance and specific in its claims. Banz and Breen (1986) showed that a standard accounting database can introduce both a look-ahead bias and an ex-post-selection bias, that results computed from it differ significantly from results computed from a database suffering from neither, and that the difference can change the conclusion of a hypothesis test. This illustrates the information-set problem the benchmark is built to measure: the bias is not merely additive noise, it can flip an inference. Kothari et al. (2005) analysed how survival requirements and data trimming affect inference, and Brown et al. (1992) and Elton et al. (1996) established how survivor-conditioned samples distort measured performance.

The as-filed-versus-restated distinction has its own literature, and it matters here because re-statement is not a uniform process. Palmrose et al. (2004) study market reactions to restatement announcements, and Hennes et al. (2008) show that restatements are heterogeneous, with errors and irregularities behaving differently. A benchmark that silently adopts the latest reported value is therefore not applying a neutral correction; it is adopting the outcome of a selective process. This is why the vintage of a financial fact is a design parameter in EDGAR-X rather than an implementation detail.

In machine learning, Kaufman et al. (2012) give the standard formulation of leakage, its detection and its avoidance. On protocol specifically, Tashman (2000) reviews out-of-sample and rolling-origin testing for forecasting, and Bergmeir and Benítez (2012) examine the use of cross-validation for time-ordered data. We read this work as establishing that protocol choice is a substantive methodological question, **not** as establishing the direction or magnitude of its effect on any particular metric, which is one of the things we measure.

Finally, the metric. At 0.73% prevalence, a summary that rewards ranking the negative class correctly is close to uninformative. Davis and Goadrich (2006) analyse the relationship between precision–recall and ROC curves, and Saito and Rehmsmeier (2015) show that the precision–recall plot is more informative than ROC for binary classifiers on imbalanced data. We therefore report average precision as the headline and ROC-AUC as secondary.

2.5 Positioning

Public bankruptcy-prediction resources differ substantially in construction. Zięba et al. (2016) apply ensemble boosted trees with synthetic feature generation over accounting features, and the accompanying data has been reused subsequently. Arno et al. (2022) introduce a benchmark for next-year bankruptcy prediction from textual data with baselines, addressing the unstructured-data setting directly.

EDGAR-X differs along three axes, stated as differences rather than as priority claims. **The information set is fixed to the filing date**, by rules that named modules enforce and continuous integration checks. **Contamination is measured rather than assumed**: we hold the benchmark fixed, vary one convention at a time, and report signed effects against decision rules fixed in advance. **Both modalities are evaluated under one protocol**, on identical test rows with one model family, which makes the modality comparison controlled.

We make no claim that EDGAR-X is the first point-in-time bankruptcy benchmark. Our search covered the sources in the reference list and was not exhaustive across the accounting, finance and machine-learning literatures. We also make no claim about how often the studies cited above do or do not apply these conventions; we did not audit their data construction.

3 The EDGAR-X Benchmark

3.1 Observational unit, cohorts and eligibility

The unit of observation is **one Form 10-K**, identified by accession number and attributed to the registrant’s **CIK**. Each choice is deliberate. The filing, not the firm-year, is what a real-time user receives. The CIK, not a ticker, is the legal-registrant identity that persists through name changes and delistings. Exact form matching to 10-K excludes 10-K/A amendments and 10-KSB, so an amendment is never silently substituted for an original.

Cohorts run **2012–2024**. The lower bound is set by XBRL coverage: structured financial data before 2012 skews toward large filers, which is the opposite of where bankruptcies concentrate. The upper bound is set by the outcome window, which requires a fully observed 365-day horizon. A 10-K is assigned to the cohort year of its **acceptance timestamp**, not its period end or filing date. Rolling evaluation tests cohort Y using only cohort years strictly below Y , leaving **2013–2024** as the twelve testable cohorts and 2012 as training-only.

An observation is eligible if its acceptance year falls inside the window and its **as-filed SIC** is known and outside the Finance, Insurance and Real Estate block (6000–6799). FIRE exclusion is a decision about accounting structure and distress mechanics rather than a “financial institution” filter, and it is applied to the code on the filing, never to a present-day code. Unknown SIC is treated as its own excluded state rather than folded into “eligible”, because that missingness is not random.

3.2 The point-in-time information protocol

The benchmark’s own inputs are load-bearing in an unusual way: leakage in the data path does not degrade the result, it *becomes* the result. The protocol is therefore a set of frozen requirements, each enforced by a named module.

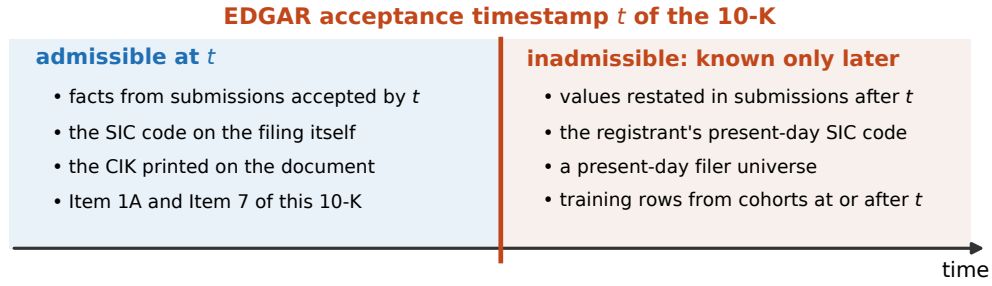
Acceptance time is the cutoff. The EDGAR acceptance timestamp, not the filing date, defines what is knowable. The two differ, and the difference is not plausibly benign in delinquent issuers, who are over-represented among bankruptcies.

Financial facts are as-filed. Data comes from the SEC DERA Financial Statement Data Sets, which are as-filed by construction: each quarterly archive is a snapshot of what was submitted

Table 1. The benchmark panel. Form 10-K observations and same-CIK initial US Chapter 7 or Chapter 11 petitions within 365 days of acceptance, 2012–2024, after FIRE and unknown-SIC exclusion. The rolling-testable population is the primary one throughout; the 2012 cohort is training-only. The two sensitivity populations were frozen before any benchmark result was observed (§5.5). Per-cohort counts are in Table A1; **the 2021 cohort carries only five positive observations** and its per-cohort figures are individually uninformative.

population	observations	positive observations	prevalence
Full panel, 2012–2024	59,500	437	0.7345%
Rolling-testable, 2013–2024 (primary)	54,351	408	0.7507%
Sensitivity: validation CIKs excluded	53,713	317	0.5902%
Sensitivity: duplicate content excluded	54,347	408	0.7507%

(a) The acceptance clock defines the information set



(b) Construction pipeline

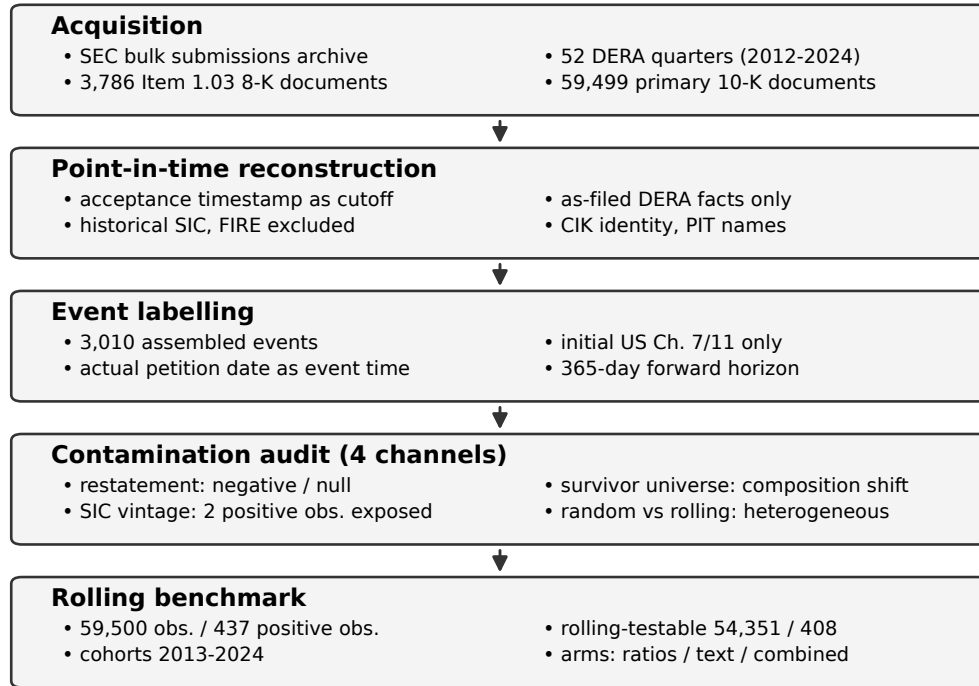


Figure 1. The EDGAR-X information-set contract and the pipeline that enforces it. (a) A schematic of the admissibility rule for one observation. The vertical line is the EDGAR acceptance timestamp t of the 10-K being scored. Everything admissible lies at or before t : financial facts from submissions accepted by then, the SIC code carried on that filing, the CIK printed on the document, and the filing’s own Item 1A and Item 7. Everything to the right is excluded by construction, including values restated later, the registrant’s present-day SIC code, a present-day filer universe, and training rows from cohorts at or after t . Panel (a) is a schematic, not a data plot. (b) Construction pipeline, read top to bottom, with the counts that survive each stage. Counts are taken from the frozen release artifacts and frozen result documents; nothing here is recomputed.

that quarter. A fact for fiscal period P may be used only if it comes from a submission accepted at or before the observation’s own acceptance timestamp; where several qualify, the most recent wins, which is what a real-time user would have had. APIs returning the *currently* reported value for a period are restatement-contaminated by design and are excluded from the data path.

Industry classification is historical. SIC is read per submission from the DERA sub table, keyed by accession. Applying a present-day code to a 2013 observation leaks a 2026 classification backwards, which is precisely the error class this study measures.

Identity is the CIK on the filing. No parent–child resolution is attempted: a subsidiary filing that the registrant is not party to is not that registrant’s event. Corporate family structure across

fifteen years of mergers and spin-offs is a research problem, not a preprocessing step. Registrant *names* are resolved point-in-time from EDGAR name history with a grace period, so a name adopted later is never used to read an earlier filing.

Event time is the petition date. Never the 8-K acceptance timestamp; substituting disclosure time would silently lengthen the horizon by the filer’s disclosure delay. Where a petition date cannot be resolved, the event is dropped rather than imputed.

3.3 Bankruptcy events and labels

An observation is **positive** if the same CIK filed an **initial US Chapter 7 or Chapter 11 petition within 365 days** of the 10-K acceptance timestamp. Each qualifier excludes a specific confusion: *initial* excludes chapter conversions; *US Chapter 7 or 11* excludes receivership, conservatorship, assignment for the benefit of creditors and foreign proceedings such as CCAA; *same CIK* excludes a subsidiary’s petition unless the registrant is demonstrably among the debtors; *filed* excludes board authorisation, preparation or stated intent.

Events are discovered from **Item 1.03** (“Bankruptcy or Receivership”) disclosures on Form 8-K, identified offline from item codes in the SEC bulk submissions archive. **3,786** Item 1.03 filings were retrieved and classified into **3,010** distinct events. Classification reads only the Item 1.03 section, because an 8-K may mention a prior Chapter 11, an affiliate proceeding or a hypothetical filing elsewhere in the document. An 8-K/A correcting a date overrides the original; explicit negation overrides affirmative-looking language elsewhere; and where identity or chapter cannot be resolved, the event is retained as **ambiguous**, an upper-bound case, rather than forced into a bucket. **One bankruptcy event can label more than one observation**, whenever two 10-K acceptances for the same registrant fall less than a horizon apart; counts of positive observations and counts of distinct events are therefore reported separately throughout. In the final panel, 437 positive observations arise from 429 distinct CIKs. Event assembly, amendment merging and the multiplicity accounting are detailed in Appendix C.

3.4 Label validation

Label quality was assessed on a **holdout of 188 cases**, specified and frozen before it was drawn, from a frame untouched at both event and CIK level, excluding any event whose accession had been inspected during development, and any event whose CIK had contributed an inspected event. Sampling was stratified with declared targets under frozen seeds, with one event per CIK to remove within-holdout dependence (Appendix D). The reviewer received an opaque case identifier, registrant name, CIK, accession numbers and the relevant filing text, and nothing else.

This holdout has been adjudicated twice, and both passes are reported because the first bears on how much weight the second carries. **First pass: AI adjudication.** The protocol specified independent human adjudication; that is not what happened first. All 188 cases were adjudicated by a **blinded AI adjudicator**. Blinding held operationally: the model saw none of the withheld fields, and the answer key was not opened until all labels were frozen, hashed and committed. **But the adjudicating model had also been involved in developing the classifier it reviewed**, so shared priors about how Item 1.03 language maps to debtor identity, chapter and petition date were common to reviewer and system under review. That pass confirmed **93 of 95** mapped positive labels: agreement 0.9789, Wilson 95% CI [0.9265, 0.9942]. **It is a classifier-to-AI-adjudicator agreement estimate, biased upward by correlated error in an unquantified way, and it is not the figure this paper relies on.**

Second pass: an independent human reference. The same frozen 188-case holdout, with the same identifiers and evidence, was re-adjudicated by human reviewers who **did not develop the classifier** and were blinded to classifier output, the AI adjudication, the answer key and the

sampling stratum. The protocol, the analysis plan and every metric below were frozen before any reviewer saw a case, as was an evidence boundary forbidding any outside lookup including an AI assistant (Appendix D). Against that human reference, **92 of the 95** mapped positive labels inside 2012–2024 are confirmed: **0.9684, Wilson 95% CI [0.9112, 0.9892]**. The denominator is fixed by the frozen sample and was not reduced. Under the frozen rule a “cannot determine” consensus counts as **not** confirmed; **none of the 95 received that consensus**, so the rule did not bind. Where the classifier commits to a specific value, agreement with the human reference is **91/92 on chapter** and **137/137 on petition date**, one case excluded as unresolvable under the consensus. Two independent first-pass reviewers, A and B, agreed on disposition in **175 of 188 cases (93.1%)**; the **binary debtor / non-debtor Cohen’s kappa is 0.872**, which is a two-category statistic and is not the coefficient corresponding to the raw multi-category 175/188 figure. On the subset where both classified the registrant as a debtor, they agreed on chapter **136/137 (99.3%)** and on petition date **137/137 (100%)**. **How the reference standard was formed, stated precisely.** The final reference therefore **equals Reviewer A’s judgements**, verified at 188 of 188 cases on all four fields. Reviewer B serves as an independent agreement check across all 188 cases, while Reviewer C, who attested independence, independently reviewed the 29 substantive disputes and selected Reviewer A’s judgement for every disputed field without supplying case-specific rationales. **This is not a conventional independently reasoned two-reviewer consensus with third-party adjudication.** A further 23 literal chapter differences, where both reviewers had already classified the registrant as a non-debtor so the chapter question did not apply and the difference was a blank against “Cannot determine”, were not routed to C; that is a **post-hoc not-applicable refinement**, recorded as such because the frozen plan carried no such carve-out. No reviewer disclosed an external lookup, so no judgement was excluded and no denominator reduced (Appendix D). The two passes reach almost the same rate on **different cases**: their non-confirmed sets are disjoint. That is consistent with the correlated-error concern motivating the second pass, and it is why the AI figure is reported as history rather than as evidence.

Nothing here relabels the benchmark. The three mapped positives the human reference does not confirm are recorded as subsidiary or affiliate filings rather than the exact registrant. They are **validation disagreements, not demonstrated label errors**: no court record was consulted, the reference standard is a human reading of the same filing text the classifier read, and the Phase 1 labels remain frozen. The two cases the AI pass disputed, a CCAA debtor in Canada whose subsidiaries filed the US Chapter 11 petitions and a Chapter 7 debtor that is a **subsidiary sharing the registrant’s exact name**, likewise survive uncorrected. Repairing any of them against the holdout would convert the holdout into development data. Any correction would require a separately declared, versioned dataset update.

3.5 Text corpus

The primary document of every frozen 10-K accession was retrieved: **59,499 of 59,500** observations (one has no primary document named in the submissions metadata), with **zero retrieval failures and zero checksum mismatches**, and all **437 positive-observation documents** obtained. Section extraction succeeded for Item 1A on **52,392** observations (88.1%) and Item 7 on **58,361** (98.1%), with both on 86.7% and at least one on 99.5%. Coverage on the positive class is higher: Item 1A 412/437, Item 7 435/437, and **at least one section for 437/437**.

The dominant Item 1A failure mode is not an extraction error: **5,788 cases are smaller reporting companies explicitly stating the item is not required**, which the extractor reports rather than passing through as substantive text. A further **179 observations (0.30% of the corpus, zero positive observations)** carry recoverable risk-factor text under a non-standard heading and are deliberately left missing under the frozen policy. Full coverage figures are in Appendix E.

4 Evaluation Design

4.1 Three feature arms

Arm A, financial ratios. Fifteen point-in-time ratios covering liquidity, leverage, profitability, cash generation and size: working capital, retained earnings, operating income, net income, revenue, equity, liabilities, cash and operating cash flow, each scaled by total assets; the current ratio; interest coverage; log assets; a negative-equity indicator; and two industry-relative terms formed against the training fold’s two-digit SIC medians. Ratios are winsorised at fixed bounds, missing components imputed with training-fold medians, and the block standardised with training-fold moments.

Arm B, filing text. Item 1A (Risk Factors) and Item 7 (MD&A) from the observation’s own 10-K. **Two separate TF-IDF vectorisers**, one per section, each capped at **100,000 features**, word unigrams and bigrams, English stop words, sublinear term frequency, minimum document frequency 5. Separate vocabularies are used because the same term means different things in the two sections: a decline discussed under Risk Factors is hypothetical, the same word in MD&A is realised. **Both vectorisers are fitted on the outer training period only**; fitting on the full corpus would leak future vocabulary and document frequencies, which is the error class under study. A missing section contributes a **zero block plus an explicit binary missingness indicator**; observations are never dropped for missing text.

Arm C, combined. Horizontal concatenation of the standardised ratio block and both text blocks with their indicators, under a single logistic regression. **No stacking, ensembling or learned gating**, no embeddings, no transformers, no LLM features, no macroeconomic variables and no additional document sections.

4.2 Model family and selection

All three arms use L2-regularised logistic regression with balanced class weights. Regularisation strength is selected from a frozen grid $C \in \{0.01, 0.1, 1.0, 10.0\}$ by an identical procedure in every arm: for test cohort Y , the inner validation window is the most recent contiguous prior cohorts jointly holding at least 25 positive observations; candidate models train only on cohorts preceding that window; the choice maximises inner-validation average precision; the selected value is refit on all observations strictly before Y . Where no such split exists, in cohorts 2013 and 2014, a fallback of $C = 1.0$ fixed in advance applies. **No test cohort and no future cohort enters selection.**

The procedure **frequently preferred the weakest regularisation in the grid**: the text and combined arms selected $C = 10$ in eight of the final ten cohorts (Appendix Table A3). We report this as an observation about the procedure on this grid, not as a claim that $C = 10$ is globally optimal; the grid was not expanded.

The three arms differ in dimensionality by four orders of magnitude, fifteen standardised ratios against up to 200,000 TF-IDF features, and we state that plainly rather than leave it to be inferred. What makes the comparison controlled is not equal capacity but equal treatment: the same regularisation grid, the same selection rule, the same training cohorts and the same test rows in every arm. The ratio arm was free to select the weakest regularisation available and did not benefit from doing so. A reader who believes the gap is a capacity effect rather than an information effect is making a testable claim, and testing it requires model families we deliberately did not run (Limitation 9).

4.3 Metrics and uncertainty

Average precision is the headline metric. At 0.73% prevalence, ROC-AUC is less informative about positive-class ranking and is therefore reported as secondary. Fixed-review-budget precision, recall and lift give the operational view. Uncertainty uses a **paired CIK-cluster bootstrap with 2,000 replicates** on identical cohort test rows, so all three arms are compared on exactly the same observations, and clustering absorbs the dependence induced by a registrant appearing in several cohorts.

Aggregates are reported as a **set of four views** (cohort median, cohort mean, event-weighted, and pooled observation-level) rather than as one number chosen after seeing the results. Event-weighted is the headline: it weights each cohort by its positive-observation count, so a five-positive cohort cannot dominate. Pooled observation-level stacks twelve differently-scaled models onto one ranking and is reported for completeness only.

Calibration is reported as a diagnostic and no arm is described as calibrated. Balanced class weighting changes the raw probability scale, most severely for the ratio arm, whose Brier scores near 0.2 reflect systematically inflated probabilities rather than poor ranking. Average precision, ROC-AUC and the review-budget metrics are the predictive comparisons; Brier score, calibration intercept and slope characterise raw model outputs only (Appendix Table A4). No post-hoc recalibration was performed.

5 Results

5.1 Filing text against financial ratios

Filing text substantially outperforms financial ratios under strict PIT evaluation. Event-weighted average precision rises from **0.0425** to **0.1301**: $\Delta\text{AP} = +0.0876$, **+206% relative, positive in 12 of 12 cohorts**, with **eight of twelve** bootstrap intervals excluding zero (2014–2018, 2020, 2021, 2023). Figure 2 shows the per-cohort series; Figure 3 shows the paired contrast with its intervals. The two weakest cohorts are 2024 (+4.1%, $P(\Delta > 0) = 0.580$) and 2022 (+66.5%, $P = 0.780$).

5.2 Does combining modalities help?

The answer depends entirely on the comparison. **Against ratios, the combined arm wins clearly:** event-weighted $\Delta\text{AP} = +0.1022$, **+240% relative, positive in 12 of 12 cohorts**, with ten of twelve intervals excluding zero (all but 2022 and 2024).

Against text alone, the picture is weaker and less consistent. Event-weighted $\Delta\text{AP} = +0.0145$, **+11.2% relative, positive in 10 of 12 cohorts**. Four cohort intervals exclude zero, and

Table 2. Aggregate average precision by arm and aggregation view, primary population, twelve test cohorts 2013–2024, 54,351 observations and 408 positive observations. All four views are reported together because the choice of aggregation is itself consequential at this sample size; every view ranks the arms in the same order. †Pooled observation-level stacks twelve differently-scaled models onto one ranking and is **not** a headline. ROC-AUC is secondary at 0.73% prevalence. Per-cohort values are in Table A1.

aggregation view	ratios	filing text	combined
cohort median	0.0289	0.0992	0.1206
cohort mean	0.0367	0.1326	0.1539
event-weighted (headline)	0.0425	0.1301	0.1447
pooled observation-level†	0.0246	0.0489	0.0769
ROC-AUC, pooled	0.7523	0.7719	0.7954

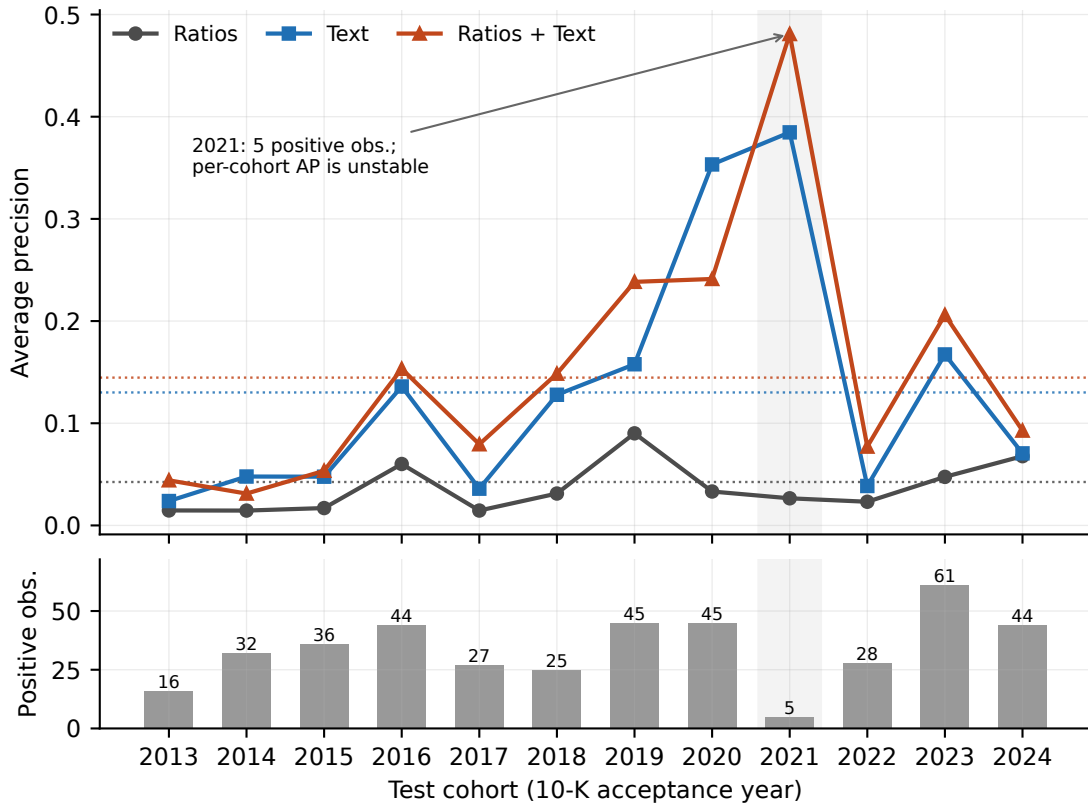


Figure 2. Average precision by test cohort, all three arms, primary population, 2013–2024. Upper panel: average precision for financial ratios (grey circles), filing text (blue squares) and the combined arm (orange triangles) on the rolling-testable population; dotted horizontal lines are each arm’s event-weighted aggregate across the twelve cohorts. Lower panel: positive observations per cohort, which are the weights. No uncertainty is shown here; per-cohort intervals for the text-over-ratios contrast are in Figure 3. The shaded 2021 column carries **five positive observations**; its level is unstable and should not be read as a finding. **Takeaway: the text and combined arms sit above the ratio arm in every cohort, and the gap is not driven by any single year.**

they do not all point the same way: **three positively** (2017, 2022, 2024) and **2020 negatively** ($\Delta AP -0.1121$, 95% CI $[-0.2037, -0.0195]$, $P(\Delta > 0) = 0.008$).

An aggregate-only report would present a clean multimodal win. The per-cohort view does not support that reading, and we do not make it. The defensible statement is that **10-K text carries substantially more bankruptcy signal than financial ratios, while ratios add only a modest and inconsistent increment beyond text.**

5.3 Analyst review budgets

Average precision is a summary of an entire ranking; an analyst reads a fixed number of filings. Figure 4 and Table 4 translate the ranking difference into that setting. **These are ranking results, not deployment claims:** no threshold, operating point or cost model is proposed, and the probabilities are not calibrated (§4.3).

At a twenty-five-filing budget the combined arm surfaces roughly three and a half times as many bankruptcy-positive filings as the ratio arm; the text-to-combined gap at these budgets is small, consistent with §5.2.

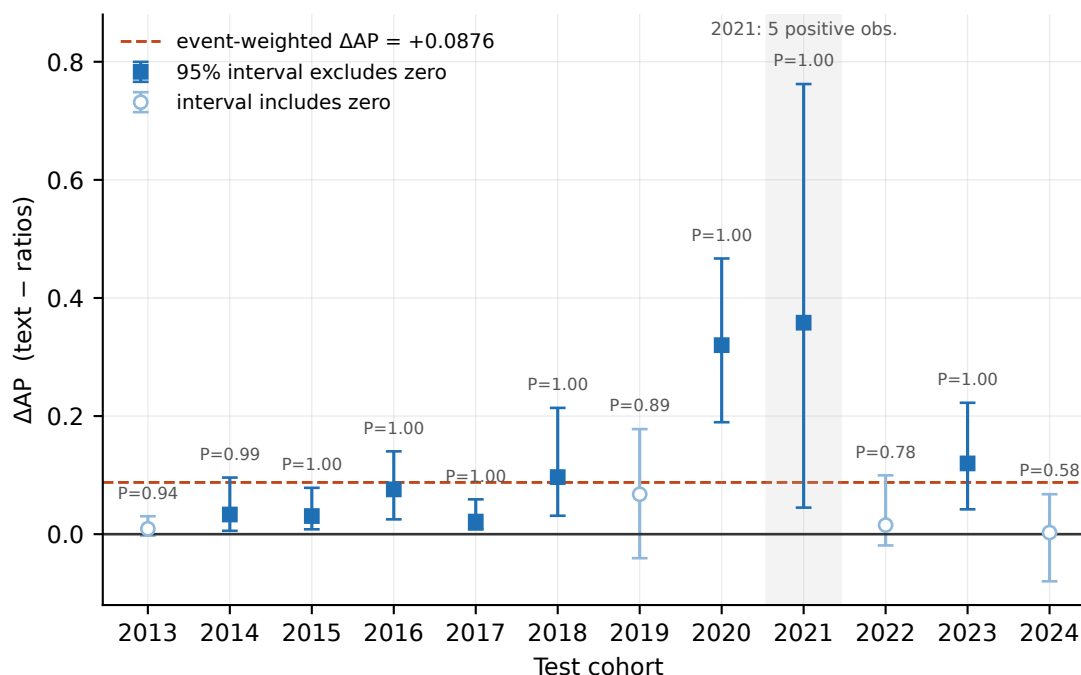


Figure 3. The text-over-ratios contrast, per cohort, with paired uncertainty. ΔAP (filing text – financial ratios) on the primary population, 2013–2024, from a paired CIK-cluster bootstrap with 2,000 replicates computed on identical cohort test rows. Filled squares mark cohorts whose 95% interval excludes zero (eight of twelve); open circles mark intervals containing zero. Error bars are 95% bootstrap intervals; P is the bootstrap $P(\Delta > 0)$. The dashed line is the event-weighted aggregate, $\Delta AP = +0.0876$. The shaded 2021 column carries five positive observations, which is why its interval is the widest in the panel. **Takeaway: the contrast is positive in every cohort, and significantly so in two-thirds of them.**

Table 3. Paired contrasts between arms, primary population, 2013–2024, paired CIK-cluster bootstrap with 2,000 replicates on identical cohort test rows. “Cohorts positive” counts the sign of the point estimate; “intervals excluding zero” counts 95% bootstrap intervals and records their direction. The single negative interval in the bottom row is 2020. Per-cohort values, with intervals and $P(\Delta > 0)$, are in Table A2. **The asymmetry between the bottom row and the two above it is the point: adding text to ratios is a large and uniform gain, adding ratios to text is neither.**

contrast	event-weighted ΔAP	relative	cohorts positive	intervals excluding zero
filing text – ratios	+0.0876	+206.1%	12 / 12	8 of 12, all positive
combined – ratios	+0.1022	+240.3%	12 / 12	10 of 12, all positive
combined – filing text	+0.0145	+11.2%	10 / 12	4 of 12: 3 positive, 1 negative

5.4 Temporal heterogeneity

Average precision rises across the panel for every arm (Figure 2), partly a training-volume effect: later cohorts train on more history. **2020 is the most influential cohort for the text-over-ratios aggregate.** Leave-one-cohort-out shows why: dropping it reduces the text advantage by about a third and doubles the combined-over-text increment (Table A5). No other cohort moves either contrast by more than roughly twelve percent, and the direction of the headline result is unchanged under every single-cohort deletion.

The 2021 cohort carries five positive observations. Its per-cohort average precision is correspondingly unstable and unusually high, and **it should not be read as a substantive finding.**

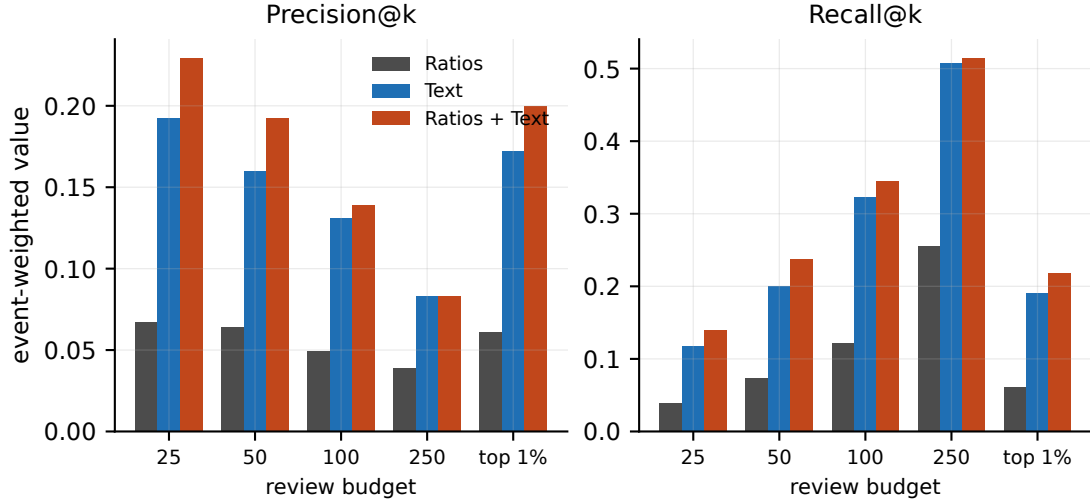


Figure 4. Performance at fixed analyst-review budgets, primary population, 2013–2024, event-weighted across the twelve cohorts. Left: precision among the top k ranked filings per cohort. Right: the share of that cohort’s positive observations captured in the top k . Budgets are 25, 50, 100 and 250 filings and the top 1% of the cohort. Lift@ k is precision@ k divided by cohort prevalence and is therefore proportional to the left panel; it appears in Table 4 rather than as a third panel. No uncertainty is shown. **Takeaway: at every budget the ordering is the same as in average precision, and the ratio-only arm is the outlier.**

Table 4. Fixed-review-budget performance, primary population, 2013–2024, event-weighted across cohorts. Precision@ k is the share of the top k ranked filings that are followed by a bankruptcy within 365 days; recall@ k is the share of the cohort’s positive observations appearing in the top k ; lift@ k is precision@ k relative to cohort prevalence. **Ranking results under a fixed review capacity, not deployment claims.** The full grid of budgets and metrics is in Table A6.

metric	ratios	filing text	combined
precision@25	0.067	0.193	0.229
lift@25	6.9×	20.6×	24.5×
recall@100	0.123	0.324	0.346
recall@250	0.255	0.507	0.515
top-1% precision	0.061	0.173	0.200

Event weighting assigns it 1.2% of the aggregate, which is the appropriate treatment; the cohort mean and median views are more exposed to it, which is one reason all four aggregation views appear together in Table 2. Figure 2 and Figure 3 shade it for the same reason.

5.5 Sensitivity analyses

Two sensitivities were frozen before any benchmark result was observed.

Validation-CIK exclusion (Option A) removes all 188 CIKs appearing in the validation hold-out, leaving 53,713 rolling-testable observations and 317 positive observations. This is the strict test of whether the holdout used for label validation contaminated the benchmark.

Duplicate-content exclusion removes four pairs of byte-identical primary documents found across distinct retrieval targets (eight documents, six CIKs, **zero positive observations**), two pairs spanning two different registrants.

Table 5. Frozen sensitivity populations, event-weighted average precision and relative contrasts, 2013–2024. Sign counts in parentheses are cohorts with a positive and negative point estimate. Both exclusions were specified before any benchmark result was observed. **Takeaway: the text-over-ratios conclusion holds in every population and in 12 of 12 cohorts in each; the combined-over-text increment is the quantity that moves, ranging from +5.5% to +17.4% and never becoming uniform across cohorts.**

Population	Obs	Pos. obs.	Event-weighted AP			Relative contrast (sign split)		
			Ratios	Text	Both	T–R	B–R	B–T
Primary (rolling-testable)	54,351	408	0.0425	0.1301	0.1447	+206.1% (12/0)	+240.3% (12/0)	+11.2% (10/2)
Validation CIKs excluded	53,713	317	0.0360	0.1079	0.1267	+199.4% (12/0)	+251.4% (12/0)	+17.4% (9/3)
Duplicate content excluded	54,347	408	0.0425	0.1355	0.1430	+218.9% (12/0)	+236.4% (12/0)	+5.5% (9/3)

6 Why Benchmark Construction Matters

The contamination audit motivates the benchmark rather than supplying its headline. It was conducted on this panel with signed effects, declared estimands and decision rules fixed before measurement, and its central message is more useful than a single correction factor would be: **contamination is not one phenomenon**. Four violations of the historical information set, routinely discussed in the same breath, change the benchmark in four different ways, and two of them cannot be expressed as a performance difference at all.

A power constraint frames every verdict. On this panel a paired design at 341 positive observations has a minimum detectable effect of roughly **24–33% relative**. Effects below that are not resolvable here, whatever their true size.

6.1 Later-restated financial facts: a negative effect

The point-in-time arm may use a fact for period P only from a submission accepted at or before the observation’s acceptance. The contaminated arm may use any submission reporting P , including one filed years later; the latest wins.

The hypothesis, fixed before measurement, was a +10–15% relative inflation in average precision. **It is not supported, and the point estimate has the opposite sign:** AP **0.0235** point-in-time against **0.0210** contaminated, Δ AP -0.0025 , **–10.49% relative**, 95% CI $[-0.0152, +0.0010]$, $P(\Delta > 0) = 0.123$. At the measured paired dependence the design has power 0.25–0.43 at +15%, so the pre-specified rule returns a kill verdict.

Restated values differ for **14.8–36.1% of positive observations** against **24.6–57.8% of negative observations**, and under contamination the positive observations’ mean rank *falls*. **This is evidence consistent with a mechanism in which failed registrants stop filing, so later revisions disproportionately land on survivors.** We do not claim the mechanism is established: it is one explanation consistent with the observed asymmetry, on one panel.

A negative result on a contaminated arm invites the reading that something is broken rather than that something is real, so the channel was ablated before the verdict was accepted. Varying one input at a time inside the contaminated arm (restated values, retroactive industry codes, and pooled rather than training-fold normalisation statistics) attributes the effect to **restated values alone**; the other two inputs do not reproduce it. The rank asymmetry above is the independent check: under contamination the positive observations’ mean rank falls while the negative observations’

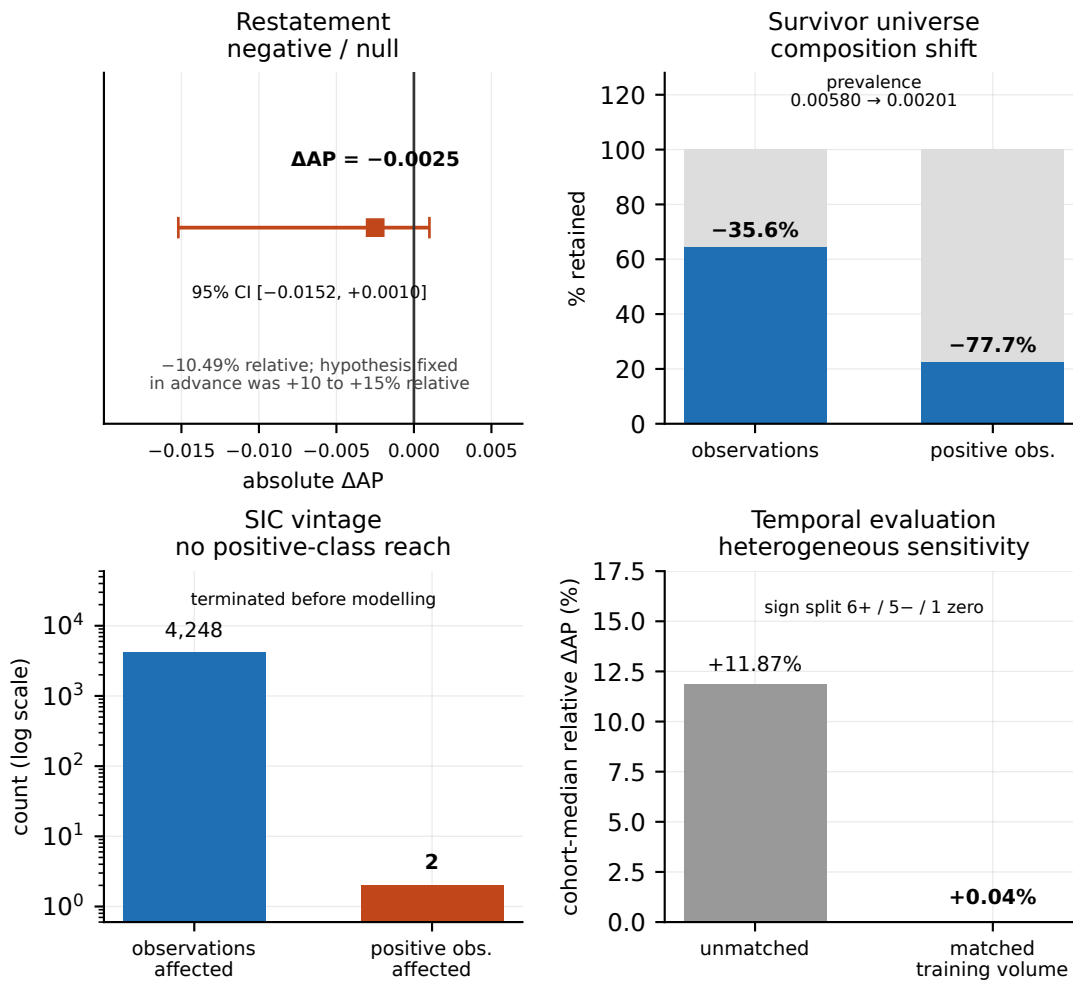


Figure 5. Four contamination channels, four different estimands. Each panel reports the quantity that channel actually admits, on the EDGAR-X panel, 2012–2024. **Panels are deliberately not on a common scale and the four numbers must not be compared or combined.** Top left: the restatement channel as a signed **absolute** ΔAP with its 95% paired-bootstrap interval, so that point estimate and interval share one scale; the relative effect and the +10 to +15% relative band fixed before measurement are annotated as text rather than plotted, because they are on a different scale. Top right: the survivor channel as composition, the share of observations and of positive observations retained under a present-day filer universe, with the resulting prevalence shift. Bottom left: the SIC-vintage channel as exposure counts on a log scale, showing that only two positive observations are affected. Bottom right: the temporal-protocol channel as the cohort-median relative ΔAP before and after holding training volume and positive-observation count fixed. **Takeaway: these are four different kinds of quantity, which is the finding.**

restatement rate is the higher of the two, which is the pattern the survivorship reading predicts and not a pattern a sign error would produce. Details are in Appendix B.

6.2 Present-day-survivor universe: a composition effect

The contaminated arm restricts the panel to registrants reachable through a present-day filer universe, operationalised as having filed anything with EDGAR in 2024. This is a stated proxy for a present-day ticker map, and a conservative one, since a delisted firm that still files counts as surviving.

The result is not a scoring effect. Observations fall from 59,500 to 37,866 and positive observations from 437 to 76: **35.6% of observations and 77.7% of positive observations removed,**

with prevalence falling from 0.00580 to 0.00201, a 65.4% reduction. We report this as counts and deliberately do not compute a powered AP effect: the paired common-intersection estimand retains only 76 positive observations, with a projected minimum detectable effect of 51–70% relative, and the composition estimand is unpaired by construction.

Two caveats attach to any use of the number. The mechanism is specifically a **2024-filer universe**, not survivorship in general; delisting, index membership, vendor back-fill and acquisition behave differently. And the filtering is strongest in early cohorts and **mechanically zero in 2024**, because a registrant that filed a 10-K in 2024 necessarily satisfies the condition, so part of the magnitude is elapsed time rather than firm mortality.

6.3 Industry-classification vintage: no positive-class reach

The contaminated arm applies the registrant’s present-day SIC retroactively, including for the FIRE eligibility test. Operational exposure is real: **4,248 observations (7.2%)** carry a changed SIC, 3,880 change two-digit group, 1,074 CIKs are affected, and 516 observations would leave the universe under today’s code. But **only two positive observations are affected**, one of them by a FIRE flip, so the channel was **terminated before any model was fitted**.

This is a statement about statistical reach, not about whether classification vintage matters operationally. It does. It simply cannot be measured as a performance effect on a panel with 437 positive observations.

6.4 Random against rolling evaluation: heterogeneous protocol sensitivity

The point-in-time arm is rolling; the contaminated arm is random observation-level 5-fold cross-validation over the pooled panel, ignoring both time and firm.

Across twelve cohorts the effect is positive in **8 of 12**, event-weighted **+2.6% relative**, 95% CI $[-0.0036, +0.0084]$. Aggregate estimators disagree by a factor of six (+2.6% event-weighted against +16.2% pooled), and the 2024 cohort alone moves the event-weighted aggregate by 7.2 percentage points. Two candidate mechanisms are **not supported as dominant**, and neither is claimed to be excluded.

Same-firm leakage. A pre-declared grouped diagnostic drove same-CIK training overlap from 79.6% to **0%** and left the effect marginally *larger* (+4.17% grouped against +2.59% ungrouped, sign preserved in 11 of 12 cohorts); the rank correlation between the effect and same-CIK exposure is +0.091. This is evidence against same-CIK overlap as the dominant mechanism on this benchmark, and it means that **CIK-grouped cross-validation, a common remedy for panel data, does not address the difference here**.

Future-population exposure. Holding training-set size and positive-observation count exactly fixed across 100 frozen matched draws per cohort, the **cohort-median relative effect collapses from +11.87% to +0.04%**, the sign split becomes 6 positive / 5 negative / 1 exactly zero, and the effect does not track future exposure (Spearman +0.175). A structural control, the final cohort, for which no future data exists and the matched pool is forced to equal the rolling training set, returns **exactly 0 with zero variance**, providing a structural check that the matching procedure does not mechanically induce a nonzero effect in the final cohort.

What remains is best described as **cohort-specific training-composition sensitivity**: *which* observations train the model matters more than *when* they are from, and the direction varies by cohort. This matched-volume diagnostic is post-hoc, motivated by the completed channel experiment; it was not fixed in advance and is not presented as confirmatory (Appendix B).

Table 6. The four contamination channels. EDGAR-X panel, 2012–2024, each channel measured against the strict point-in-time benchmark with one convention varied at a time. **Estimands are not commensurable and the rows must not be combined into a single “cost of contamination”.** Where both are reported, the confidence interval is on the absolute Δ AP scale and the percentage is a relative point estimate; no relative interval was computed. Panel-wide minimum detectable effect is approximately 24–33% relative at 341 positive observations. Protocol details, including the decision rules fixed before measurement and the declared deviations, are in Appendix B.

channel	estimand	result	verdict
Later-restated financials	signed paired Δ AP; interval on the absolute scale	AP 0.0235 PIT vs 0.0210 contaminated; absolute Δ AP = -0.0025 , 95% CI $[-0.0152, +0.0010]$; relative point estimate -10.49%	Inflation hypothesis fixed in advance not supported ; sign is negative
Present-day-survivor universe	composition, not performance	-35.6% observations, -77.7% positive observations; prevalence 0.00580 $\rightarrow$ 0.00201	Reported as counts; paired estimand retains 76 positive observations (MDE 51–70%)
Industry-classification vintage	signed absolute Δ AP on affected subset	4,248 observations (7.2%), 1,074 CIKs, 2 positive observations affected	Terminated before modelling for lack of positive-class reach
Random vs rolling evaluation	signed paired Δ AP; interval on the absolute scale	8/12 cohorts positive; relative point estimate $+2.6\%$ event-weighted, absolute 95% CI $[-0.0036, +0.0084]$; CIK grouping does not remove it; matched volume collapses the cohort median from $+11.87\%$ to $+0.04\%$ relative	Heterogeneous protocol sensitivity, not stable inflation

6.5 What the audit establishes

No single contaminated convention on this panel produced a stable, powered performance-inflation effect. That is a bounded claim, not a licence: at 341–437 positive observations the panel cannot resolve contamination effects at the $\pm 10\%$ relative scale, so these are bounds on dominance rather than evidence of absence. The finding that does generalise is about method: a study reporting a precise contamination effect on a panel of this size should state its minimum detectable effect.

7 Discussion

Why might filing narrative dominate conventional ratios? The empirical result is that it does, under a strict information set, by a factor of three in event-weighted average precision. Two interpretations are consistent with the data and neither is tested here. The first is informational: Item 1A and Item 7 are where management discusses going-concern doubt, covenant pressure, refinancing risk and liquidity in specific language, while a ratio block summarises the same balance sheet into fifteen numbers that are, by construction, also available to every other firm in the sector. The second is about specificity: a distressed firm’s disclosure names its own circumstances, whereas ratio thresholds must generalise across very different capital structures. We label both as interpretation. What the experiment establishes is the comparison, not its cause.

Both arms are restricted to the same acceptance timestamp, and the vectorisers are fitted on the training period alone, so the text arm sees no future vocabulary and no future document frequencies. Those two leakage routes are closed by construction. We do not claim the gap is free of every possible artifact of the text pipeline, only of the ones the protocol prevents.

Why do ratios add only a modest increment once text is present? The most economical reading is redundancy: if the ratio block’s signal is largely a summary of conditions that manage-

ment also describes in prose, then the marginal contribution of ratios given text should be small, which is what we observe, +11.2% event-weighted. The 2020 cohort, where adding ratios makes things significantly worse, suggests something further, namely that the ratio block can mislead under a macroeconomic shock that affects balance sheets broadly without distinguishing the firms that fail. That is speculation about one cohort and we flag it as such. The reportable finding is the heterogeneity itself.

What does the audit teach about financial ML evaluation? Three things. First, contamination channels should be named and measured individually, because they do not share an estimand: a composition change and a scoring change are not the same kind of object, and averaging them produces a number that means nothing. Second, grouping cross-validation folds by firm, a common remedy for panel data, does not address the rolling-versus-random difference on this benchmark: it removes essentially all same-firm overlap and leaves the measured difference slightly larger. Whatever drives that difference here, it is not the mechanism that remedy targets. Third, the direction of a contamination effect cannot be assumed. For the restatement channel, the hypothesis fixed before measurement was inflationary; on this panel its point estimate is negative, with evidence consistent with a survivorship mechanism that would make the negative sign unsurprising in hindsight.

What will EDGAR-X enable? The benchmark is prepared for release with its protocol, its frozen results and its reproduction commands, and it is constructed so that a future study can vary one convention while holding everything else fixed. The obvious extensions are ones we deliberately did not run: stronger text representations under the same information-set contract; longer or shorter horizons; and the excluded FIRE sector under its own accounting conventions. The independent human adjudication that v1 named as the most valuable outstanding work has since been carried out and is reported in §3.4; what remains outstanding there is a linkage study against an external court record, which is the only route to an accuracy measurement rather than an agreement one, and to any statement about label recall. The methodological contribution is the contract, not the model.

On power. That this panel resolves a 200% modality effect and not a 10% contamination effect is a fact about the problem, not a flaw in the design: reportable bankruptcies among SEC registrants are rare.

8 Limitations

1. **Validation is agreement with a human reference, not ground truth.** The headline validation figure, 92 of 95 mapped positives confirmed, is agreement between the classifier and a human reading of the same filing text the classifier read. No court record, docket or external database was consulted, so it is **not** an accuracy measurement against an independent record of what was filed. It measures label precision on discovered events and nothing else.
2. **The human reference is one reviewer's judgement, ratified rather than independently re-reasoned.** Reviewer C selected Reviewer A for every disputed field without supplying case-specific rationales, so the reference equals Reviewer A's judgements in full. Reviewer B functions as an independent agreement check, not as a contributor to the standard. This is weaker than the two-reviewer consensus with independently reasoned adjudication that the protocol described, and stronger than the earlier AI adjudication by an adjudicator that had helped build the classifier. Both the earlier pass and its limitation are retained in §3.4 as study history.
3. **The bankruptcy definition is narrow by construction.** Same-CIK initial US Chapter 7 or Chapter 11 within 365 days. Receivership, conservatorship, assignment for the benefit

of creditors, foreign proceedings, subsidiary-only petitions and chapter conversions are not positives. A different definition would produce a different benchmark.

4. **Label recall is not measurable from inside EDGAR.** Bankruptcies that Item 1.03 never reports cannot be detected without an external reference such as the Federal Judicial Center’s Integrated Database (Federal Judicial Center) or the Florida-UCLA-LoPucki Bankruptcy Research Database (University of Florida Levin College of Law). Neither is a drop-in solution: the BRD covers only large public companies, requiring a Form 10-K or Form 10 filer with assets of \$100 million or more in 1980 dollars, and has not been updated since its December 2022 update. **What we measure is label precision on the events we find.**
5. **Five mapped positives are disputed by one adjudication pass or the other and all remain in the benchmark:** the CCAA case and the same-name-subsidiary case flagged by the AI pass, and three the human reference records as subsidiary or affiliate filings. The two sets are disjoint. None is a demonstrated label error, and all are deliberately uncorrected so the holdout was not converted into development data.
6. **Statistical power.** With 437 positive observations at 0.73% prevalence the panel cannot resolve contamination effects at the $\pm 10\%$ relative scale. Null results in §6 bound dominance, not existence.
7. **Sparse cohorts.** The 2021 cohort carries five positive observations; its per-cohort figures are individually uninformative and are shaded in Figures 2 and 3 for that reason. Cohort mean and median views are more exposed to it than the event-weighted headline.
8. **Known extraction limitations.** 179 Item 1A sections (0.30% of the corpus, zero positive observations) are recoverable in principle and left missing under the frozen policy. Filings that abandon item numbering entirely have no terminator and are structurally undefined for section extraction.
9. **A sparse linear text model, by design.** TF-IDF with logistic regression was chosen for reproducibility and to keep the information-set contract auditable. Modern neural text representations may change effect sizes; they would not change the protocol, and we did not test them.
10. **Probabilities are not calibrated deployment probabilities.** Balanced class weighting changes the raw probability scale, most severely for the ratio arm. Calibration statistics in Table A4 are diagnostics of raw outputs. No threshold, operating point or cost model is proposed anywhere in this paper.
11. **The survivor channel uses one specific definition,** filed anything with EDGAR in 2024, with intrinsic cohort truncation that makes it mechanically zero in the most recent cohort. It is not a general result about survivorship.
12. **Scope, and what the scope costs.** US SEC registrants filing Form 10-K in English, 2012–2024, with Finance, Insurance and Real Estate excluded on the as-filed code. Two boundaries carry a real cost and we state it rather than only its rationale. The 2012 floor, set by XBRL coverage, excludes the 2008–2009 financial crisis, so this benchmark is silent about that crisis regime. The FIRE exclusion is a choice about accounting structure, not a data constraint, and it removes a sector in which distress is both common and economically consequential. Results do not transfer automatically to private firms, non-US jurisdictions or the excluded sectors.

9 Conclusion

EDGAR-X fixes every input of a bankruptcy benchmark against one clock, the EDGAR acceptance timestamp of the filing being scored, and enforces that contract in code rather than in prose.

On it we ask what violating the historical information set actually does, and what a model can learn when the information set is respected.

The audit’s answer is that contamination is not a single quantity. Four channels that are routinely discussed together behave differently in direction, magnitude and estimand: one negative, one a change in the benchmark’s composition rather than its scores, one untestable for lack of positive-class exposure, and one a heterogeneous protocol sensitivity that firm-grouped cross-validation does not address on this benchmark. Reporting a single “cost of contamination” would misrepresent all four.

The central empirical finding is about modality. Under strict point-in-time evaluation, **10-K narrative text carries substantially more bankruptcy signal than financial ratios**: event-weighted average precision 0.1301 against 0.0425, a 206% relative improvement positive in all twelve cohorts, robust to two frozen sensitivities and to every single-cohort deletion, **while ratios add only a modest and heterogeneous increment beyond text**. The comparison holds under a common sparse linear model family applied identically to both arms; we did not test whether a higher-capacity model narrows it. For a task that has been studied through accounting ratios for six decades, that is a notable place to find the signal. Within this benchmark and sparse linear model family, most of the observed predictive advantage lies in what management wrote.

The benchmark, its protocol, its frozen results and its reproduction commands are publicly available at <https://github.com/bhavyaupadhyay/edgar-x-benchmark>. The raw SEC corpus is not redistributed; it is reacquirable from primary sources with the included tooling.

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Appendices

A Complete tables

These tables carry the per-cohort detail behind §5. Every value is read directly from the frozen release artifacts in `research/bankruptcy_benchmark/gate4_release/` by `paper/tables/build_tables.py`; nothing is transcribed by hand and nothing is recomputed from predictions.

Table A1. Complete per-cohort results, primary population, 2013–2024.

Cohort	Pos. obs.	Average precision			ROC-AUC		
		Ratios	Text	Both	Ratios	Text	Both
2013	16	0.0147	0.0239	0.0441	0.8316	0.8706	0.8932
2014	32	0.0145	0.0479	0.0312	0.6815	0.8125	0.7962
2015	36	0.0170	0.0477	0.0535	0.7481	0.8100	0.8586
2016	44	0.0601	0.1360	0.1533	0.7831	0.9024	0.8785
2017	27	0.0146	0.0360	0.0794	0.6940	0.8605	0.8510
2018	25	0.0312	0.1279	0.1485	0.8366	0.8995	0.9234
2019	45	0.0901	0.1577	0.2384	0.7540	0.8739	0.8475
2020	45	0.0332	0.3533	0.2412	0.7971	0.9390	0.9469
2021	5	0.0266	0.3847	0.4809	0.9009	0.9965	0.9951
2022	28	0.0232	0.0386	0.0771	0.7338	0.7867	0.8287
2023	61	0.0475	0.1673	0.2060	0.7191	0.8708	0.8750
2024	44	0.0678	0.0705	0.0928	0.7221	0.8579	0.8817
Cohort median		0.0289	0.0992	0.1206			
Cohort mean		0.0367	0.1326	0.1539			
Event-weighted		0.0425	0.1301	0.1447			
Pooled observation-level		0.0246	0.0489	0.0769	0.7523	0.7719	0.7954

Note. Pooled observation-level stacks twelve differently-scaled models onto one ranking and is not a headline. ROC-AUC is secondary at 0.73% prevalence. The 2021 cohort carries five positive observations.

Table A2. Complete per-cohort paired comparisons, primary population, 2013–2024.

Contrast	Cohort	Δ AP	Relative	95% CI	$P(\Delta > 0)$
Text – Ratios	2013	+0.0092	+62.7%	[-0.0025, +0.0304]	0.942
	2014	+0.0333	+229.2%	[+0.0057, +0.0959]	0.995
	2015	+0.0307	+180.6%	[+0.0082, +0.0785]	0.998
	2016	+0.0759	+126.2%	[+0.0251, +0.1403]	0.998
	2017	+0.0214	+146.8%	[+0.0080, +0.0590]	1.000
	2018	+0.0967	+309.7%	[+0.0313, +0.2139]	0.999
	2019	+0.0676	+75.0%	[-0.0407, +0.1780]	0.893
	2020	+0.3201	+964.2%	[+0.1896, +0.4669]	1.000
	2021	+0.3581	+1348.8%	[+0.0450, +0.7623]	1.000
	2022	+0.0154	+66.5%	[-0.0191, +0.0997]	0.780
	2023	+0.1198	+252.1%	[+0.0421, +0.2225]	1.000
	2024	+0.0027	+4.1%	[-0.0798, +0.0677]	0.580
	Text – Ratios event-wtd	+0.0876	+206.1%	median +0.0505 / mean +0.0959	12+/0–
Both – Ratios	2013	+0.0295	+200.9%	[+0.0058, +0.1030]	0.999
	2014	+0.0166	+114.3%	[+0.0051, +0.0401]	0.999
	2015	+0.0365	+214.4%	[+0.0155, +0.0805]	1.000
	2016	+0.0932	+155.0%	[+0.0417, +0.1773]	1.000

Contrast	Cohort	Δ AP	Relative	95% CI	$P(\Delta > 0)$
	2017	+0.0648	+444.9%	[+0.0198, +0.1907]	1.000
	2018	+0.1173	+375.7%	[+0.0432, +0.2600]	1.000
	2019	+0.1482	+164.5%	[+0.0579, +0.2442]	1.000
	2020	+0.2080	+626.5%	[+0.1127, +0.3330]	1.000
	2021	+0.4543	+1711.1%	[+0.0303, +0.9419]	1.000
	2022	+0.0539	+232.8%	[-0.0046, +0.1490]	0.953
	2023	+0.1585	+333.5%	[+0.0750, +0.2572]	1.000
	2024	+0.0250	+36.9%	[-0.0525, +0.1050]	0.816
Both – Ratios	event-wtd	+0.1022	+240.3%	median +0.0790 / mean +0.1172	12+/0–
Both – Text	2013	+0.0203	+84.9%	[-0.0004, +0.0878]	0.972
	2014	-0.0167	-34.9%	[-0.0655, +0.0045]	0.061
	2015	+0.0058	+12.1%	[-0.0168, +0.0299]	0.687
	2016	+0.0173	+12.7%	[-0.0128, +0.0708]	0.845
	2017	+0.0435	+120.8%	[+0.0040, +0.1435]	0.988
	2018	+0.0206	+16.1%	[-0.0097, +0.0737]	0.925
	2019	+0.0806	+51.1%	[-0.0084, +0.1795]	0.960
	2020	-0.1121	-31.7%	[-0.2037, -0.0195]	0.008
	2021	+0.0962	+25.0%	[-0.0467, +0.3555]	0.680
	2022	+0.0385	+99.8%	[+0.0003, +0.0745]	0.977
	2023	+0.0386	+23.1%	[-0.0101, +0.0883]	0.945
	2024	+0.0223	+31.6%	[+0.0058, +0.0574]	0.997
Both – Text	event-wtd	+0.0145	+11.2%	median +0.0214 / mean +0.0212	10+/2–

Note. Paired CIK-cluster bootstrap, 2,000 replicates, seed 20260916, on identical cohort test rows. Text – Ratios excludes zero in 8 of 12 cohorts; Both – Ratios in 10 of 12; Both – Text in 3 of 12 positively (2017, 2022, 2024) with 2020 excluding zero negatively.

Table A3. Selected regularisation strength C by population and test cohort.

Cohort	Option A			Primary (B)			Duplicate-excluded		
	Ratios	Text	Both	Ratios	Text	Both	Ratios	Text	Both
2013	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
2014	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
2015	0.01	1.0	1.0	0.01	10.0	10.0	0.01	1.0	10.0
2016	0.01	0.1	10.0	0.01	1.0	10.0	0.01	1.0	10.0
2017	0.1	0.01	0.1	10.0	0.01	0.1	0.1	0.01	0.1
2018	0.1	1.0	0.1	0.01	10.0	10.0	0.01	10.0	10.0
2019	0.01	10.0	10.0	10.0	10.0	0.1	10.0	10.0	0.1
2020	0.01	0.01	0.1	10.0	10.0	0.1	10.0	10.0	0.1
2021	10.0	1.0	10.0	1.0	10.0	10.0	1.0	10.0	10.0
2022	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
2023	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
2024	0.01	10.0	10.0	0.01	10.0	10.0	0.01	10.0	10.0

Note. Selected by the frozen procedure on prior cohorts only. 2013 and 2014 take the $C = 1.0$ fallback fixed in advance. The procedure frequently preferred the weakest regularisation in the frozen grid; this is not a claim that $C = 10$ is globally optimal.

Table A4. Calibration diagnostics of raw model outputs, primary population.

Cohort	Brier score			Cal. intercept			Cal. slope		
	Ratios	Text	Both	Ratios	Text	Both	Ratios	Text	Both
2013	0.21639	0.00579	0.00709	-5.812	-2.722	-2.870	0.659	1.007	0.984
2014	0.19707	0.00755	0.00855	-4.913	-1.414	-2.280	0.420	1.104	0.790
2015	0.21934	0.00889	0.00844	-4.908	-2.115	-1.789	0.767	0.548	0.644
2016	0.23073	0.01340	0.00984	-4.868	-1.771	-1.316	1.059	1.113	0.769
2017	0.22186	0.15962	0.05911	-5.057	-4.551	-3.950	0.573	2.358	1.083
2018	0.22116	0.00578	0.00560	-5.403	-0.695	-0.609	1.193	0.965	0.979
2019	0.21112	0.01017	0.04797	-4.625	-0.797	-3.080	0.978	0.823	1.258
2020	0.22328	0.00986	0.06774	-4.606	-1.079	-3.582	0.914	0.997	1.677
2021	0.21441	0.00242	0.00332	-7.377	-2.509	-3.074	1.391	1.325	1.215
2022	0.20294	0.00615	0.00604	-5.029	-1.536	-1.106	0.717	0.498	0.630
2023	0.23120	0.01372	0.01361	-4.318	-1.209	-1.269	0.677	0.686	0.713
2024	0.23785	0.01291	0.01355	-4.662	-1.809	-1.843	0.789	0.675	0.700

Note. **No arm is described as calibrated.** Balanced class weighting changes the raw probability scale, most severely for the ratio arm. These are diagnostics of raw frozen logistic outputs only; no recalibration was performed.

Table A5. Leave-one-cohort-out influence on the event-weighted aggregates.

Dropped	Event-weighted AP			Contrast	
	Ratios	Text	Both	Text – Rat.	Both – Text
none	0.0425	0.1301	0.1447	+0.0876	+0.0145
2013	0.0436	0.1345	0.1488	+0.0908	+0.0143
2014	0.0449	0.1371	0.1543	+0.0922	+0.0172
2015	0.0450	0.1381	0.1535	+0.0931	+0.0154
2016	0.0404	0.1294	0.1436	+0.0890	+0.0142
2017	0.0445	0.1368	0.1493	+0.0923	+0.0125
2018	0.0432	0.1303	0.1444	+0.0870	+0.0141
2019	0.0366	0.1267	0.1330	+0.0901	+0.0063
2020	0.0437	0.1025	0.1327	+0.0588	+0.0302
2021	0.0427	0.1270	0.1405	+0.0843	+0.0135
2022	0.0439	0.1369	0.1496	+0.0929	+0.0128
2023	0.0416	0.1236	0.1339	+0.0820	+0.0103
2024	0.0395	0.1373	0.1509	+0.0979	+0.0136

Note. **2020 is the most influential cohort.** The direction of the headline result is unchanged under every single-cohort deletion.

Table A6. Complete fixed-review-budget grid, event-weighted across cohorts.

Budget k	Metric	Ratios	Text	Both
25	precision	0.0672	0.1925	0.2294
	recall	0.0392	0.1176	0.1397
	lift	6.88×	20.60×	24.46×
50	precision	0.0641	0.1601	0.1928
	recall	0.0735	0.2010	0.2377
	lift	6.47×	17.67×	20.98×
100	precision	0.0494	0.1314	0.1389
	recall	0.1225	0.3235	0.3456
	lift	5.40×	14.33×	15.27×
250	precision	0.0392	0.0829	0.0834
	recall	0.2549	0.5074	0.5147
	lift	4.56×	9.03×	9.17×
top 1%	precision	0.0610	0.1725	0.1998
	recall	0.0613	0.1912	0.2181
	lift	6.10×	19.06×	21.76×

Note. **Ranking and analyst-review results, not deployment claims.** No threshold, operating point or cost model is proposed.

Table A7. Per-cohort panel statistics, rolling-testable population.

Test cohort	Observations	Positive obs.	Prevalence
2013	5,048	16	0.3170%
2014	5,060	32	0.6324%
2015	4,990	36	0.7214%
2016	4,553	44	0.9664%
2017	4,492	27	0.6011%
2018	4,350	25	0.5747%
2019	4,240	45	1.0613%
2020	4,111	45	1.0946%
2021	4,251	5	0.1176%
2022	4,575	28	0.6120%
2023	4,456	61	1.3689%
2024	4,225	44	1.0414%
2013–2024	54,351	408	0.7507%

Note. Full panel 2012–2024: 59,500 observations, 437 positive observations, prevalence 0.7345%. The 2012 cohort is training-only under rolling evaluation. **2021 carries only five positive observations**; its per-cohort figures are individually uninformative.

B Contamination audit protocol

Specification before measurement, and what that does and does not mean. All four channels were specified with a declared estimand, a signed hypothesis, a decision rule and a minimum detectable effect, and those specifications were committed to this repository before any result was observed; the protocol document was committed on 2026-09-16 and the channel results on 2026-09-20, so the precedence is checkable from the commit history. **Nothing was lodged with an external preregistration registry, and no claim of external preregistration is made anywhere in this paper.** The specifications are frozen in GATE2B_PROTOCOL.md; the

results in GATE2A_RESULT.md, GATE2B_CHANNEL3_RESULT.md, GATE2C_RESULT.md and CONTAMINATION_FINDINGS_FREEZE.md.

Estimands. Channels 1 and 4 admit a paired signed Δ AP on a common intersection of observations, reported with a confidence interval on the absolute scale and a relative point estimate; no relative interval was computed. Channel 2 does not: restricting to a present-day filer universe changes which rows exist, so the paired estimand retains only 76 positive observations and the composition estimand is unpaired by construction. It is therefore reported as counts. Channel 3 was terminated on exposure before any model was fitted. **The four quantities are not commensurable and were never combined.**

Power. The panel-wide minimum detectable effect is approximately 24–33% relative at 341 positive observations under the measured paired dependence. The restatement channel has power 0.25–0.43 against the +15% alternative fixed in advance. The survivor channel’s paired estimand has a projected minimum detectable effect of 51–70% relative. These figures are the reason several results are stated as bounds on dominance rather than as evidence of absence.

Declared deviations. Two are material and are stated wherever the relevant result appears. First, the label-validation protocol initially deviated from plan when the first pass used a blinded AI adjudicator rather than independent humans. A subsequent second pass used independent human reviewers under the frozen human-validation protocol; both passes and their limitations are reported in §3.4 and Appendix D. Second, the matched-volume diagnostic in §6.4 is **post-hoc**: it was motivated by the completed temporal-protocol experiment and is recorded as a mechanism diagnostic in GATE2C_MATCHED_VOLUME_PROTOCOL.md, explicitly not as confirmatory analysis fixed in advance. The 2024 structural control was included in that diagnostic as a structural check on the matching procedure, and returned exactly zero with zero variance.

Channel ablation. The restatement result is attributed to restated values rather than to retroactive industry codes or pooled normalisation statistics by an ablation that varies one input at a time within the contaminated arm.

C Event assembly and multiplicity

Amendment merging. An original 8-K and its amendments describe one bankruptcy under different accession numbers, so deduplicating on accession double-counts them. Events are clustered within a CIK by identical docket number; by matching chapter with petition dates within a small tolerance; or by an actual /A filing within an amendment window of a still-incompletely-identified event. **The proximity rule is restricted to amendments.** Applying it to ordinary 8-Ks would merge two genuine bankruptcies, which is a worse error than splitting one. The negative merge guards in the benchmark’s integration suite exist so that a later loosening of this rule fails a test.

Multiplicity. One bankruptcy can label more than one observation whenever two 10-K acceptances fall less than a horizon apart. This is not a rare edge case: for a delinquent filer catching up on two years of 10-Ks two months apart the duplication window is roughly 305 days wide, and delinquent catch-up filing concentrates in exactly the distressed population that supplies the positive class. Multiplicity is therefore correlated with the label. The benchmark reports **unique events and positive observations separately**: in the final panel, 437 positive observations arise from 429 distinct CIKs, so within-panel duplication is modest. The accounting is kept explicit because selecting only the latest 10-K per registrant would quietly redefine the scientific target.

Ambiguous outcomes. Where debtor identity or chapter cannot be resolved from the filing, the event is retained as ambiguous and treated as an upper-bound case rather than forced into a bucket.

D Validation methodology

Frame. The holdout was drawn from a frame untouched at both event and CIK level: excluding any event whose accession had been inspected during development, **and** any event whose CIK had contributed any inspected event. The second condition is the stricter one and is what makes Option A in §5.5 a meaningful sensitivity.

Sampling. 188 cases, stratified with declared targets across mapped Chapter 11, mapped Chapter 7, ambiguous, subsidiary-only, receivership-only, foreign-proceeding-only and prospective-only strata, under frozen seeds, with review order randomised under a separate seed. An amendment before adjudication enforced one event per CIK to remove within-holdout dependence.

Blinding. The reviewer received an opaque case identifier, registrant name, CIK, accession numbers, the Item 1.03 text, amendment text and nearby filing text. The automated disposition, automated petition date, sampling stratum and all development annotations were withheld. The answer key was not opened until all reviewer labels were frozen, hashed and committed.

First pass, and its deviation. Adjudication was first performed by a blinded AI adjudicator rather than the independent human reviewers the protocol specified, and that adjudicator had also been involved in developing the classifier under review. The provenance is in FINAL_AI_ADJUDICATION_PROVENANCE.md and the deviation in Amendment 2 of FINAL_VALIDATION_PROTOCOL.md. That pass confirmed 93 of 95 mapped positives: 0.9789, Wilson 95% CI [0.9265, 0.9942], CIK-weighted 0.9823, with chapter and petition-date agreement 93/93. Every figure from it is a classifier-to-AI-adjudicator agreement estimate and is retained here as study history, not as evidence. **Second pass: independent human adjudication.** The same frozen 188 cases, the same opaque identifiers and the same evidence were re-adjudicated by human reviewers who did not develop the classifier and were blinded to classifier output, the AI adjudication, the answer key and the sampling stratum. The protocol and the full analysis plan, including every metric and the treatment of “cannot determine”, were frozen before any reviewer saw a case. Two dated amendments, also pre-review, fixed the evidence boundary, which forbids any web, EDGAR, court-record, repository, AI-assistant or third-party lookup, and the handling of a disclosed lookup. Reviewer eligibility excluded anyone who had worked on the classifier, and no AI system was permitted as a reviewer or as a disagreement resolver. Two reviewers, A and B, independently labelled all 188 cases without communicating. Twenty-nine substantive disagreements went to a third reviewer, C, who attested independence and selected Reviewer A’s judgement for every disputed field without supplying case-specific rationales; the final reference therefore equals Reviewer A’s judgements, verified at 188 of 188 cases on all four fields. A further 23 literal chapter differences arose where both reviewers had already classified the registrant as a non-debtor, so the chapter question did not apply and the difference was a blank against “Cannot determine”; these were not routed to C, a post-hoc not-applicable refinement recorded as such because the frozen plan carried no such carve-out. No reviewer disclosed an external lookup, so no judgement was excluded and no denominator reduced. **Second-pass results.** 92 of 95 mapped positives confirmed: 0.9684, Wilson 95% CI [0.9112, 0.9892], denominator fixed by the frozen sample and not reduced; no mapped positive drew a “cannot determine” consensus. Classifier against the human reference: chapter 91/92, petition date 137/137 with one case excluded as unresolvable. First-pass reviewer agreement: disposition 175/188 (93.1%), binary debtor / non-debtor Cohen’s kappa 0.872, chapter 136/137 and petition date 137/137 on the subset both classified as a debtor. The AI pass and the human pass flag disjoint sets of non-confirmed cases. The protocol, analysis plan, amendments, reviewer packets, sealed key, analysis script and every reported number are in `research/bankruptcy_benchmark/human_validation/`, with the results package under `results_phase2a/`.

Secondary error audit. An audit over non-positive strata is reported in the frozen result documents. It is explicitly **not** an in-window recall estimate, and its ambiguous-stratum intervals are too narrow because six of its hits are six separate registrants in a single bankruptcy docket.

E Text retrieval and extraction coverage

Retrieval. 59,499 of 59,500 observations have a primary document named in the submissions metadata; the one exception was never requested. All 59,499 were retrieved with **zero failures and zero SHA-256 mismatches**, including all 437 positive-observation documents.

Table A8. Section extraction coverage, EDGAR-X panel 2012–2024. Positive-class coverage exceeds corpus-wide coverage for every section.

section	observations	share	positive obs.	share
Item 1A (Risk Factors)	52,392	88.1%	412 / 437	94.3%
Item 7 (MD&A)	58,361	98.1%	435 / 437	99.5%
both sections	51,571	86.7%	410 / 437	93.8%
at least one section	n/a	99.5%	437 / 437	100%

Failure modes. The dominant Item 1A gap is not an extraction error: **5,788 cases are smaller reporting companies explicitly stating the item is not required**, which the extractor reports as such rather than passing through as substantive text. A further **179 observations (0.30%, zero positive observations)** carry recoverable risk-factor text under a non-standard heading and are deliberately left missing under the frozen policy; recovering them after the extractor was frozen would have made the corpus a function of inspection. Filings that abandon item numbering entirely have no terminator and are structurally undefined for section extraction.

Missing-text policy. A missing section contributes a zero block plus an explicit binary missingness indicator. No observation is dropped for missing text, so the evaluation population is identical across all three arms.

Cache equivalence. The text cache used by the benchmark was verified byte-equivalent to the frozen extractor’s output across all 59,500 observations, with zero mismatches, before any benchmark model was trained.

F Label policy summary

Table A9. Label policy. The full policy with its rationale is frozen in LABEL_POLICY.md. Each row excludes a specific confusion observed in the Item 1.03 corpus rather than stating a general preference.

dimension	policy
identity	CIK on the filing; no parent–child resolution
chapter	initial US Chapter 7 or Chapter 11 only
conversions	not a new event
excluded proceedings	receivership, conservatorship, assignment for the benefit of creditors, CCAA, administration, winding-up
subsidiary petitions	not the registrant’s event unless the registrant is demonstrably among the debtors
intent	board authorisation, preparation or stated intention is not an event
event time	actual petition date; never the 8-K acceptance timestamp
unresolvable dates	event dropped, never imputed
amendments	8-K/A correcting a date overrides the original
negation	explicit negation overrides affirmative-looking language elsewhere
ambiguity	retained as an upper-bound case, not forced into a bucket
horizon	365 days from the 10-K acceptance timestamp

G Reproducibility

What the release contains. The public release is available at <https://github.com/bhavyaupa-dhyay/edgar-x-benchmark>. It comprises the benchmark code, the frozen protocol documents, the release artifacts in `gate4_release/` (per-cohort metrics, aggregate metrics, selected regularisation strengths, leave-one-cohort-out results and an artifact manifest), and the commands that reproduce every table and figure in this paper. `paper/tables/build_tables.py` and `paper/figures/*.py` read only those release artifacts.

What the release excludes. The raw SEC corpus and the extracted text cache are not redistributed. They are reacquirable from primary sources with the included tooling, which enforces the SEC’s rate limit and requires a contact address on the command line.

Frozen seeds. The paired CIK-cluster bootstrap uses 2,000 replicates under seed 20260916. Validation sampling, review ordering and the matched-volume draws each use their own frozen seeds, recorded in the corresponding protocol document.

Determinism. The regularisation grid, the inner-validation rule, the extraction policy and the evaluation population were all frozen before the benchmark was run. The benchmark was executed once; no result in this paper is a re-run.

Provenance of every number. `paper/CLAIM_AUDIT.md` maps each numerical claim in this manuscript to the frozen artifact that supports it.

Frozen commits. The empirical study is frozen at `dcee1037bcb21e358be531d988e147fd7641e696`, with a dated prose erratum at `377d57bead56053786ef7d062bc43a3ebaea5f57` correcting one interval tally. No result in this paper was recomputed after those commits.